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CASE STUDY

Topic : A Case Study on Improving Production Workflow at Swift Cycle Works

S.No	Description	Maximum Mark	Marks Awarded
1.	Problem Understanding	10	
2.	Innovation & Creativity	20	
3.	Presentation, Communication & Response for Queries	10	
	Total Marks	**40**	

Subject Incharge

1. Title

Optimization of Assembly Line Efficiency: A Comprehensive Case Study on Improving Production Workflow, Eradicating Bottlenecks, and Implementing Lean Methodologies at Swift Cycle Works

2. Executive Summary

This comprehensive operations management case study investigates the production inefficiencies currently paralyzing Swift Cycle Works, a mid-sized bicycle manufacturing enterprise. Faced with a sudden 66% surge in market demand (scaling from 300 to 500 units daily), the organization attempted to meet targets through brute-force methods, specifically mandatory prolonged overtime and accelerated manual labor pacing. This stopgap approach resulted in severe operational degradation, yielding only 350 units daily alongside a 15% spike in product defect rates, heightened machinery breakdowns, and critical workforce fatigue.

Through rigorous on-site empirical data collection—including time-motion studies, machinery downtime logs, ergonomic movement mapping, and qualitative employee interviews—this study diagnosed a critical bottleneck at Station 3 (Gears and Brakes Installation), which recorded a cycle time of 12 minutes compared to the 3-to-5-minute averages of adjacent stations. Furthermore, an absence of standardized preventive maintenance and chaotic workstation layouts exacerbated downtime.

The recommendations formulated herein bypass capital-intensive automation in favor of highly effective, low-cost Lean Manufacturing principles. By rebalancing the assembly line workload (shifting specific tasks from Station 3 to Station 2), implementing the 5S methodology for spatial organization, establishing proactive maintenance schedules, and cross-training the workforce, Swift Cycle Works can achieve its 500-unit daily target within standard operating hours. This document provides a complete theoretical framework, granular data analysis, financial impact assessment, and a phased implementation roadmap to guide the facility back to sustainable profitability.

3. Introduction

3.1 The Macro-Environment of Manufacturing

Manufacturing companies in the contemporary global economy operate under relentless pressure to achieve simultaneous and often conflicting goals: maximizing throughput, minimizing operational costs, and strictly adhering to elevated quality standards. The core mechanism driving manufacturing success is the assembly line—a sequential, highly

synchronized production process where interchangeable parts are added to a semi-finished product in a logical progression. The efficacy of an assembly line dictates the financial health of the organization. When optimized, it generates a steady flow of high-margin goods; when mismanaged, it becomes a sinkhole for capital, labor hours, and brand reputation.

3.2 Introduction to Swift Cycle Works

This case study centers on Swift Cycle Works, a fictional yet highly representative medium-sized manufacturing facility specializing in standard urban commuter bicycles. Historically, Swift Cycle Works enjoyed a stable market presence, operating a predictable, single-shift assembly line optimized for a daily output of 300 bicycles. However, macro-environmental shifts over the past two years—specifically, urban infrastructure investments in bike lanes and a cultural shift toward eco-friendly commuting—triggered an unprecedented surge in customer demand.

3.3 The Catalyst for Operational Crisis

Initial managerial responses to the demand surge were optimistic but operationally flawed. To meet the newly established target of 500 bicycles per day, facility managers relied exclusively on extending labor hours and demanding faster human output without altering the fundamental physics of their assembly line layout. The original line architecture was fundamentally incapable of processing 500 units per shift without structural rebalancing.

Currently, the factory is in a state of crisis. Output has plateaued at approximately 350 units per day despite exorbitant overtime expenditures. More alarmingly, the strain on the system has manifested as a sharp decline in product quality, with customers reporting loose pedals, uncalibrated brakes, and cosmetic damage. The factory floor is visually chaotic, the workforce is physically and morally exhausted, and mechanical assets are failing due to overuse and neglect. This study unpacks the systemic failures at Swift Cycle Works and applies academic operations management principles to engineer a sustainable turnaround.

4. Theoretical Framework and Literature Review

To properly analyze and correct the deficiencies at Swift Cycle Works, it is necessary to ground the investigation in established Operations Management paradigms.

4.1 The Theory of Constraints (ToC)

Introduced by Eliyahu M. Goldratt, the Theory of Constraints postulates that every complex system or manufacturing process consists of multiple linked activities, one of which acts as a constraint or "bottleneck." The throughput of the entire system is strictly dictated by the

capacity of this single bottleneck. Improving the speed of non-constraint operations does not increase overall output; it merely creates excess work-in-progress (WIP) inventory. Identifying and elevating the constraint is the only mathematical way to increase systemic throughput.

4.2 Lean Manufacturing and the Reduction of 'Muda'

Originating from the Toyota Production System (TPS), Lean Manufacturing focuses on the relentless elimination of waste, termed *Muda*. Waste in this context is defined as any activity that consumes resources but creates no value for the end customer. Lean identifies several types of waste present at Swift Cycle Works:

- **Motion:** Unnecessary movement of workers (e.g., walking to find tools).
- **Waiting:** Idle time experienced by workers downstream of a bottleneck.
- **Defects:** Products requiring rework or causing customer returns.
- **Over-processing:** Using inappropriate tools or procedures.

4.3 The 5S Methodology

A foundational component of Lean, 5S is a systemic framework for workplace organization designed to improve efficiency and safety:

1. **Sort (Seiri):** Eliminate all unnecessary tools, parts, and debris from the workspace.
2. **Set in Order (Seiton):** Arrange essential items logically so they can be easily accessed.
3. **Shine (Seiso):** Clean the workspace and equipment continuously.
4. **Standardize (Seiketsu):** Create uniform procedures for the first three S's.
5. **Sustain (Shitsuke):** Build a culture of discipline to maintain the standards.

5. Problem Statement

Swift Cycle Works is experiencing a severe, multi-faceted degradation in assembly line performance, characterized by an inability to meet a critical production quota of 500 units per day (currently stalled at 350 units). This 30% production deficit is directly resulting in delayed shipments, breached vendor contracts, and deteriorating customer satisfaction.

The organizational attempt to bridge this capacity gap via mandatory, sustained overtime has proven economically unviable and physically detrimental to the workforce. Consequently, worker fatigue and a disorganized production environment have triggered a 15% escalation in manufacturing defects over the preceding financial quarter. The core problematic phenomena include a severely unbalanced workflow architecture, an absence of proactive asset maintenance leading to unpredictable work stoppages, and a lack of spatial standardization on the factory

floor. Without immediate operational intervention, the company faces severe financial losses and irrecoverable brand damage.

6. Objectives and Scope of the Study

6.1 Primary Objectives

The central aim of this case study is to conduct a granular analysis of the Swift Cycle Works assembly process and deploy operations management interventions to achieve the 500-unit daily target without relying on overtime or sacrificing quality.

6.2 Secondary Objectives

- **Bottleneck Diagnostics:** To empirically isolate the specific workstation(s) throttling the flow of production using time-motion analysis.
- **Workload Equalization (Line Balancing):** To redistribute assembly sub-tasks across available workstations, ensuring equitable cycle times and minimizing idle waiting periods.
- **Asset Reliability Optimization:** To transition the facility from a reactive (break-fix) maintenance posture to a Total Productive Maintenance (TPM) framework.
- **Ergonomic and Spatial Reorganization:** To eliminate wasted motion and physical strain by implementing 5S principles and point-of-use tooling.
- **Quality Assurance:** To drastically reduce the 15% defect rate by standardizing part selection and reducing the cognitive load on stressed operators.
- **Workforce Welfare:** To eliminate the necessity for mandatory overtime, thereby reducing occupational fatigue, absenteeism, and worker turnover.

6.3 Scope and Limitations

The scope of this study is restricted to the physical assembly line operations within the primary Swift Cycle Works facility. It does not encompass upstream supply chain logistics (raw material procurement) or downstream distribution channels (shipping logistics).

7. Research Methodology

To guarantee a robust and scientifically sound analysis, a mixed-methods approach was utilized over a two-week observation period on the factory floor.

7.1 Quantitative Data Collection

- **Time and Motion Studies:** Researchers utilized digital stopwatches to record cycle times at each of the five workstations. A sample size of 50 consecutive assembly cycles was measured to calculate accurate mean task durations and identify variations.

- **Downtime Audits:** Historical maintenance logs from the past 30 days were digitized and categorized to quantify the exact hours of production lost to specific machinery failures.
- **Defect Tracking:** Quality control logs were analyzed to categorize the types of defects (e.g., cosmetic vs. mechanical) and trace them back to specific workstations.

7.2 Qualitative Data Collection

- **Ergonomic Observation:** Researchers mapped the physical pathways of operators, noting instances of bending, reaching, and walking away from the primary workspace.
- **Structured Interviews:** Brief, 10-minute structured interviews were conducted with 20 assembly line workers across all five stations to capture subjective data regarding fatigue, process frustration, and morale.

8. Comprehensive Data Collection and Observations

The empirical data gathered during the two-week diagnostic phase revealed glaring systemic flaws.

8.1 Observation 1: Cycle Time Tracking and Station Analysis

The assembly line is linearly segmented into five primary workstations. The following table represents the calculated mean time required to complete the designated tasks for a single bicycle unit.

Station Number	Task Description	Average Cycle Time (Minutes)	Variance Observed
Station 1	Frame preparation, degreasing, and painting	4.0	Low (Consistent)
Station 2	Hub alignment, adding wheels, and mounting tires	3.0	Low (Consistent)
Station 3	Installing complex gear cassettes and brake systems	**12.0**	**High (Bottleneck)**
Station 4	Mounting saddle (seat), stem, and handlebars	4.0	Low (Consistent)

Station Number	Task Description	Average Cycle Time (Minutes)	Variance Observed
Station 5	Final torque inspection, manual testing, and boxing	5.0	Medium (Depends on defects)

Total Unbalanced Cycle Time: 28 Minutes per unit. However, due to the sequential nature of the line, the pace is strictly governed by Station 3, meaning a new bicycle can only exit the line every 12 minutes.

8.2 Observation 2: Machinery Uptime and Maintenance Logs

An audit of the previous month's mechanical logs highlighted a complete absence of preventive care.

- **Thermal Curing Unit (Station 1):** The paint-drying heater suffered critical failures 4 times in one month. Because it lacked regular sensor calibration, it overheated and triggered safety shutdowns, costing a total of 10 hours of line stoppage.
- **Pneumatic Torque Wrenches (Station 3):** The automatic screwdrivers jammed repeatedly—averaging 15 to 20 minutes of micro-stoppages per daily shift. This was traced directly to a lack of daily lubrication and clearing of metal shavings.
- **Maintenance Posture:** The facility operates on a "Run-to-Failure" model. Maintenance technicians are only dispatched *after* an operator raises a red flag, leading to cascading delays.

8.3 Observation 3: Motion Waste and Ergonomic Hazards

Spatial analysis mapping revealed severe inefficiencies in how workers interact with their environment.

- **Tool Hunting (Station 3):** The worker responsible for gears and brakes must walk approximately 10 steps to a communal toolbox to retrieve specific Allen keys and wrenches for different bicycle models. This action occurs an estimated 350 times a day, equating to roughly 3,500 wasted steps per shift.
- **Ergonomic Strain (Station 2):** Raw materials (heavy rubber tires and metal rims) are stored on the concrete floor on wooden pallets. The operator must perform a deep lumbar bend 350 times a day to retrieve these parts.

- **Inventory Chaos:** Consumables (screws, bolts, washers, and cables) are stored in mixed, unlabelled cardboard boxes on the workstations.

8.4 Observation 4: Human Capital and Workforce Feedback

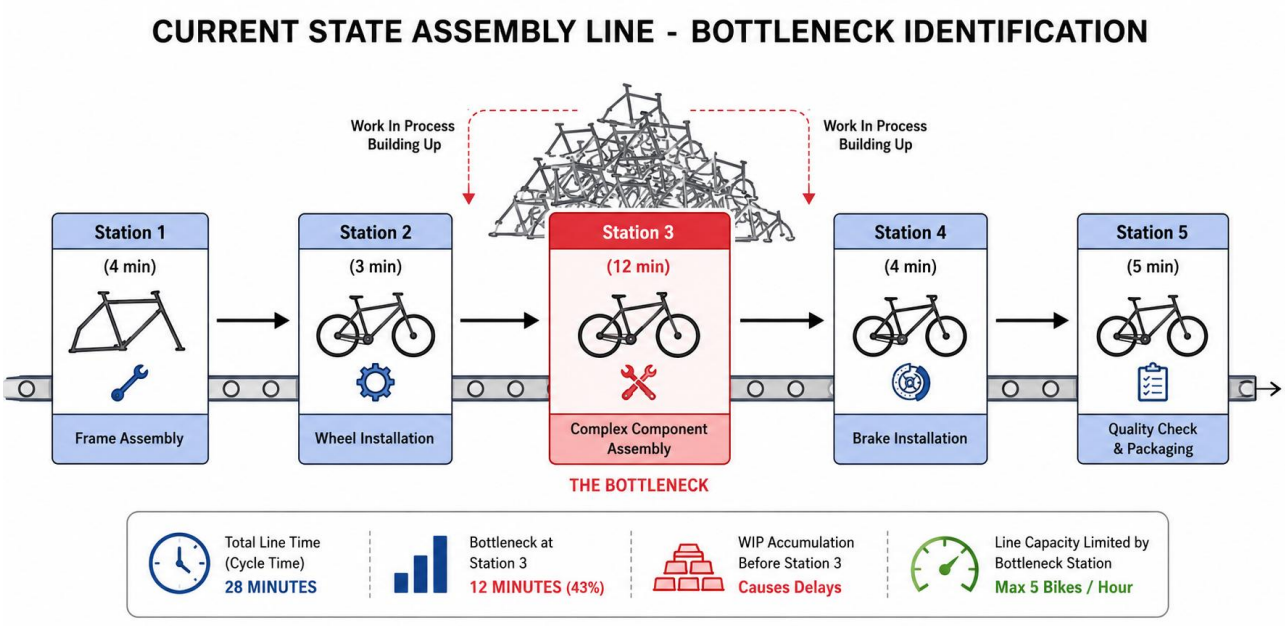
Qualitative interviews highlighted a deeply fractured team dynamic caused by the unbalanced line.

- **Idle Frustration (Stations 2 & 4):** Workers expressed boredom and guilt. Because their tasks take only 3-4 minutes, they spend up to 8 minutes per cycle standing idle, waiting for Station 3 to finish.
- **Extreme Stress (Station 3):** The operators at the bottleneck station reported severe psychological stress. They feel rushed, are prone to panic, and feel unfairly blamed by management for holding up the entire factory.
- **Physical Exhaustion:** 85% of interviewed staff reported chronic back pain and sleep deprivation stemming from mandatory 10-to-12-hour shifts.

9. In-Depth Analysis and Root Cause Identification

Utilizing the collected data, we can synthesize the root causes of the operational failures and map their cascading impacts on the enterprise.

9.1 Root Cause 1: The Severe Bottleneck at Station 3



The fundamental law of factory physics is that a system's capacity is equal to the capacity of its bottleneck. Station 3 takes 12 minutes. Therefore, the theoretical maximum output of this

line, assuming zero breakdowns and zero breaks, is exactly 5 units per hour (60 mins / 12 mins). In an 8-hour shift, this yields only 40 bicycles. Even with 4 hours of overtime (12-hour shift), the maximum theoretical output is 60 bicycles—far short of the 500 daily target (which would require multiple parallel lines, or a drastically reduced cycle time). Station 3 is overloaded with two highly complex, precision-heavy tasks: threading delicate derailleur cables (gears) and calibrating tension (brakes).

9.2 Root Cause 2: Amplification of Waste via Poor Layout

The layout of the factoryfloor is actively working against the employees. In Lean terminology, the motion waste observed at Station 3 is catastrophic. The time spent walking to a communal toolbox is non-value-added time. When compounded over hundreds of cycles, this translates to hours of lost production. Furthermore, the chaotic storage of mixed fasteners directly causes the quality defects; a stressed worker at Station 3, rushing to catch up, reaches into a mixed bin and utilizes a slightly undersized bolt on a brake pad, leading to the customer complaints documented in Section 3.

9.3 Root Cause 3: The Reactive Maintenance Trap

Operating machinery without routine maintenance is a false economy. By saving a few minutes in the morning by skipping lubrication, the company loses hours in the afternoon to jammed tools. When Station 1's heater breaks down, the entire line is starved of painted frames. Stations 2, 3, 4, and 5 must halt completely until the heater is fixed.

9.4 Financial and Reputational Impacts

- **Margin Erosion:** Paying time-and-a-half wages for overtime while output remains stagnant is devastating to the company's profit margins. Furthermore, the cost of processing returns, shipping replacement parts, and scrapping defective frames adds a massive financial burden.
- **Labor Turnover:** Burnout is imminent. If the current conditions persist, experienced workers will resign, forcing the company to spend capital on recruiting and training novices, which will further slow down the line.
- **Brand Devaluation:** In the era of e-commerce, consumer reviews dictate market share. Continuing to ship defective units will permanently tarnish Swift Cycle Works' reputation, ensuring that when the current bicycle boom subsides, the company will have no loyal customer base to sustain it.

10. Summary of Findings

Based on the exhaustive diagnostic process, the following critical findings are formalized:

1. **Critical Line Imbalance:** The assembly line is constrained by a severe bottleneck at Station 3 (12-minute cycle time), which dictates the throughput of the entire facility and forces upstream/downstream stations into non-productive idle states.
2. **Rampant Motion Waste:** Disorganized tool storage and floor-level material staging force workers into repetitive, time-consuming motions that add zero value to the product and cause cumulative physical trauma.
3. **Unreliable Asset Infrastructure:** A reactive-only maintenance philosophy is causing highly predictable, easily preventable machinery breakdowns that halt the entire value stream.
4. **Quality-Speed Tradeoff:** The 15% surge in product defects is a direct symptom of cognitive overload and disorganized materials at the bottleneck station, rather than inherent worker incompetence.
5. **Ineffective Management Interventions:** The application of mandatory overtime as a blanket solution to a structural flow problem has entirely failed, succeeding only in exhausting the workforce and eroding profit margins.

11. Strategic Recommendations and Implementation Plan

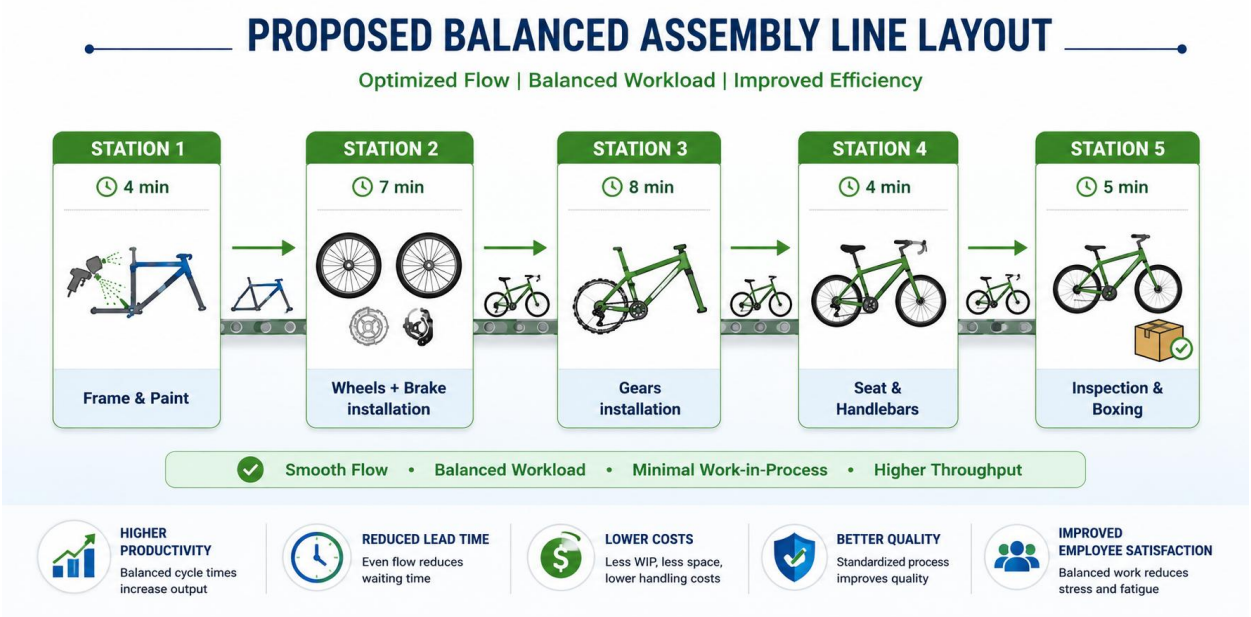
To rectify the systemic failures at Swift Cycle Works, management must pivot away from "working harder" and immediately transition to "working smarter." The following operations management strategies must be implemented sequentially.

11.1 Recommendation 1: Assembly Line Rebalancing (Workload Equalization)

The immediate priority is to break the bottleneck at Station 3. This requires a redistribution of labor elements.

- **Task Shifting:** The installation of brakes and the installation of gears are two separate sub-assemblies currently grouped together. Management must detach the "Brake Installation" process from Station 3 and reassign it to Station 2 (which currently only takes 3 minutes).
- **New Cycle Times:** By shifting this labor, Station 2 will see an increase in cycle time to approximately 7 minutes. Station 3, relieved of the brake assembly, will drop from 12 minutes to approximately 8 minutes.

- **Impact:** The line will now flow significantly smoother. The gap between the fastest and slowest stations shrinks dramatically, reducing idle time and allowing products to move continuously without piling up.



11.2 Recommendation 2: Implementation of the 5S Methodology

The physical workspace must be re-engineered to support human operators, not hinder them.

- **Sort & Set in Order (Point-of-Use Tooling):** Abolish the communal toolbox. Install custom pegboards directly at eye level at every single workstation. Every tool must have a designated, outlined shadow on the board. The worker should be able to reach any tool without moving their feet.
- **Ergonomic Material Staging:** Purchase hydraulic lift tables or simple raised platforms for Station 2. Heavy tires and rims must be presented to the worker at waist height. This eliminates the 350 daily lumbar bends, preserving worker energy for late-shift productivity.

- **Standardize (Visual Management):** Implement a Kanban bin system. Screws, bolts, and washers must be separated into distinct, color-coded bins (e.g., Red for brakes, Blue for gears). This visual cue completely eliminates the cognitive burden of searching and prevents the usage of incorrect parts, immediately solving the defect crisis.



11.3 Recommendation 3: Transition to Total Productive Maintenance (TPM)

The culture of "Run-to-Failure" must end.

- **Autonomous Maintenance:** Operators must be given ownership of their machines. Station 3 workers should be trained to perform a daily 5-minute cleaning and lubrication of their pneumatic wrenches before the shift begins.
- **Preventive Scheduling:** The maintenance department must implement a strict calendar. The thermal curing unit at Station 1 must have its sensors calibrated and heating coils inspected every Friday after the final shift, ensuring 100% uptime during the working week.

11.4 Recommendation 4: Cross-Training and Workforce Flexibility

A rigid workforce creates fragility in the system.

- **Skill Matrix:** Develop a training matrix mapping every employee's competency at every station.
- **Rotation:** Implement a mandatory rotation where employees spend one day a month working a different station. If the primary painter at Station 1 is absent, the line must not stop; a cross-trained worker from Station 4 should immediately step in. This also alleviates the boredom reported by workers at faster stations.

11.5 Recommendation 5: Abolition of Mandatory Overtime

With the line balanced and wastes eliminated, the physical capacity of the line will increase naturally.

- **Restoration of Morale:** Suspend all mandatory overtime immediately. Re-establish a standard 8-hour shift to allow workers to recover physically and mentally.
- **Incentivization:** Replace the stick with a carrot. Establish a team-based bonus structure. If the newly balanced line achieves the 500-unit target within the standard 8-hour shift with less than a 1% defect rate, the entire shift receives a financial bonus. This aligns worker goals with company goals.

12. Change Management and Risk Mitigation

Implementing operational changes often faces resistance from a workforce accustomed to the status quo, even if the status quo is painful.

- **Communication:** Management must hold a town hall meeting to explain *why* these changes are happening. They must admit that the overtime strategy was flawed and frame the 5S and rebalancing efforts as tools to make the workers' lives easier.
- **Pilot Testing:** Before locking in the new line balance, run it as a "pilot program" for three days. Solicit feedback from the workers at Station 2 and 3 to fine-tune the exact tool placements and task hand-offs.
- **Monitoring:** The Operations Manager must spend the first two weeks physically present on the floor, coaching workers through the new 5S standards to ensure they don't slip back into old habits (the 'Sustain' phase of 5S).

13. Conclusion

The operational crisis at Swift Cycle Works serves as a textbook example of the limitations of brute-force management in a complex manufacturing environment. Faced with a sudden surge in demand, the instinctive reaction to demand more hours and faster hands from the

workforce precipitated a catastrophic drop in efficiency and quality. Through meticulous data collection and the application of proven operations management principles, it became evident that the root cause was not a lack of effort, but a fundamental lack of systemic optimization.

The presence of a severe bottleneck, combined with egregious motion waste and reactive maintenance, artificially capped the factory's potential. By stepping back and applying Lean methodologies—specifically line rebalancing to equalize cycle times, 5S visual management to organize the physical space, and preventive maintenance to secure asset reliability—Swift Cycle Works can achieve a radical transformation.

These interventions require minimal capital expenditure but demand a high degree of managerial discipline and organizational culture change. By executing the recommendations detailed in this study, Swift Cycle Works will not only meet its critical target of 500 bicycles per day, but will do so within standard operating hours, restoring both their financial profit margins and the morale of their vital human workforce. This case underscores the enduring truth of operations management: sustainable productivity is achieved by engineering a better process, not by exhausting the people within it.

14. References

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